

EXPERIMENT NO.06REALISATION OF FLIP FLOPS USING NAND GATESAim :

Realise flipflops using NAND gates.

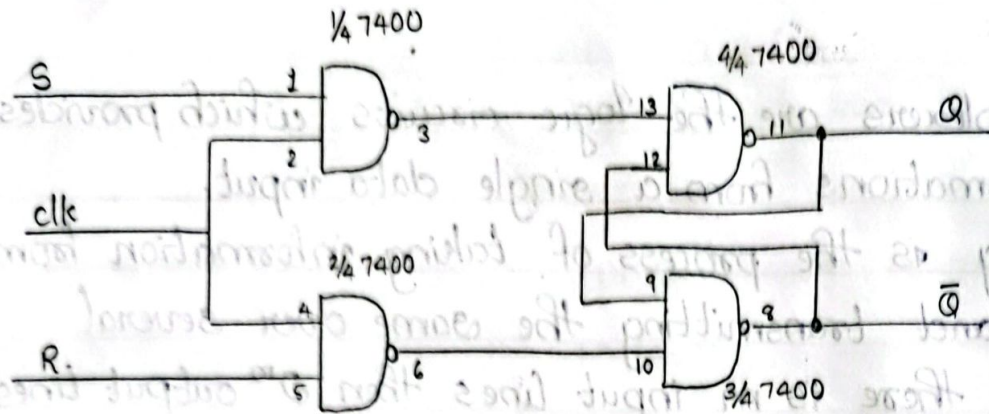
Components Used :

S.No.	Components	Specification	Quantity
1	IC Trainer kit		1
2	Connecting wires		1
3	IC test kit		1
4	IC 7400	Quad 2 i/p NAND gate	2
5	IC 7404	Hex Inverter	1
6	IC 7410	Triple 3 i/p NAND gate	1

Procedure :

The simplest type of flipflop is called a SR latch. It has 2 o/p's labelled Q and $\bar{Q}$ and 2 i/p's labelled S and R . The state of the latch components to the level of a (high or low) 1000 and $\bar{Q}$ is of course, the complement of that state. It can be constructed using either two even coupled NAND gate or two coupled NOR gate. Using two NOR gate an active high SR latch can be constructed and using two NAND gates or active low SR latch can be constructed using either two even coupled NAND gates or Active low SR can be constructed. The name of the latch SR or set - reset is derived from the name of d's i/p's.

SR Flip Flop



Circuit Diagram

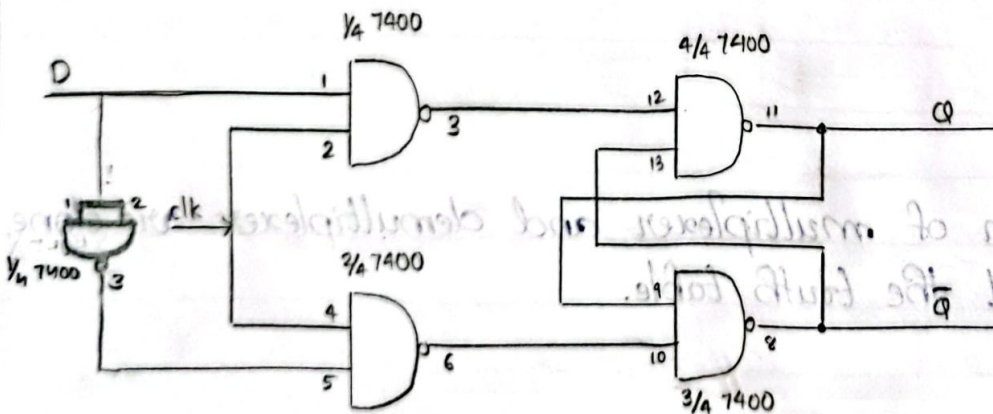


Logic Symbol

S	R	Q_n	State
0	0	Q_n	No Change
0	1	0	Reset
1	0	1	Set
1	1	X	-

Truth Table

D Flip Flop



Circuit Diagram



Logic Symbol

D	Q_n	State
0	0	Reset
1	1	Set

Truth Table

D-Flipflop

In many applications it is not necessary to have separate S and R inputs to a latch of the input combination $S=R=0$ and $S=R=1$ are never needed. The S and R are always the complement of each other. So we can construct a latch with a single input and obtained the R input by converting it. The single input is labelled D and the device is called a D latch.

J-K Flipflop

The JK flip flop is a very versatile and also the most widely used. J and K designations for synchronous control inputs have no known significance. The functioning of JK flip flop is identical to that of the SR flip flop except that it has no invalid state like that of the SR flip flop.

T Flipflop

A T flip flop has a single control input labelled T for toggle when T is HIGH. The flip flop toggles on every new clock pulse when T is low. The flip flop remains in whatever state it was before. Although T-flip flop remains in whatever state it was before. Although T flip flops are not widely available commercially it is easy to convert a JK flip flop to the functional equivalent of a T flip flop by just connecting J and K together and labelling the common connection as T. This when $T=1$, we have $J \cdot K = 1$ and flip flop toggles.

IMPLEMENTATION OF THE FLIP FLOP USING GATES

Use flip-flops using logic gates

IC 7410

IC 7410

IC 7410

IC 7410

IC 7410

IC 7410

IC 7410

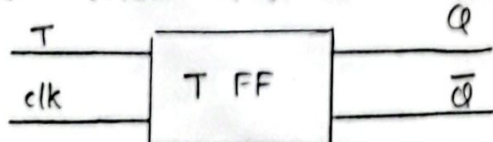
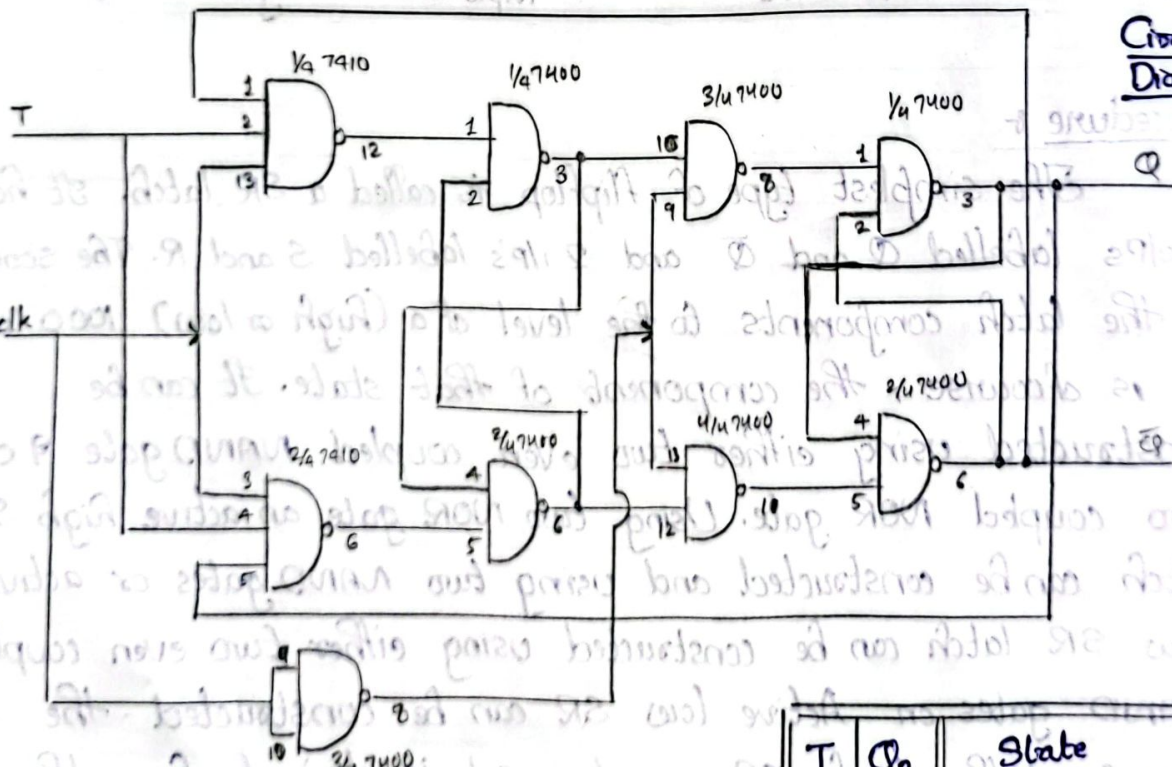
IC 7410

IC 7410

IC 7410

T-Flip Flop Master Slave

Circuit Diagram



Logic Symbol

T	Q _n	State
0	Q _n	No change
1	Q _n	Toggle

Master slave J-K Flipflop

The truth table operation is the same as that of a negative edge triggered JK flip-flop except for the way, in which it is triggered. The logic diagram of master-slave JK flipflop is similar to that of the master slave SR flipflop. The master-slave JK flipflop is a JK flipflop followed by a SR flipflop. The problem of logic race is eliminated, because when the clock drives the master, the inverted clock drive the slave.

PROCEDURE

1. Check the IC's given
2. Connect the IC as per the circuit
3. Check the connections
4. Give the supply
5. Check the op obtained with Truth table

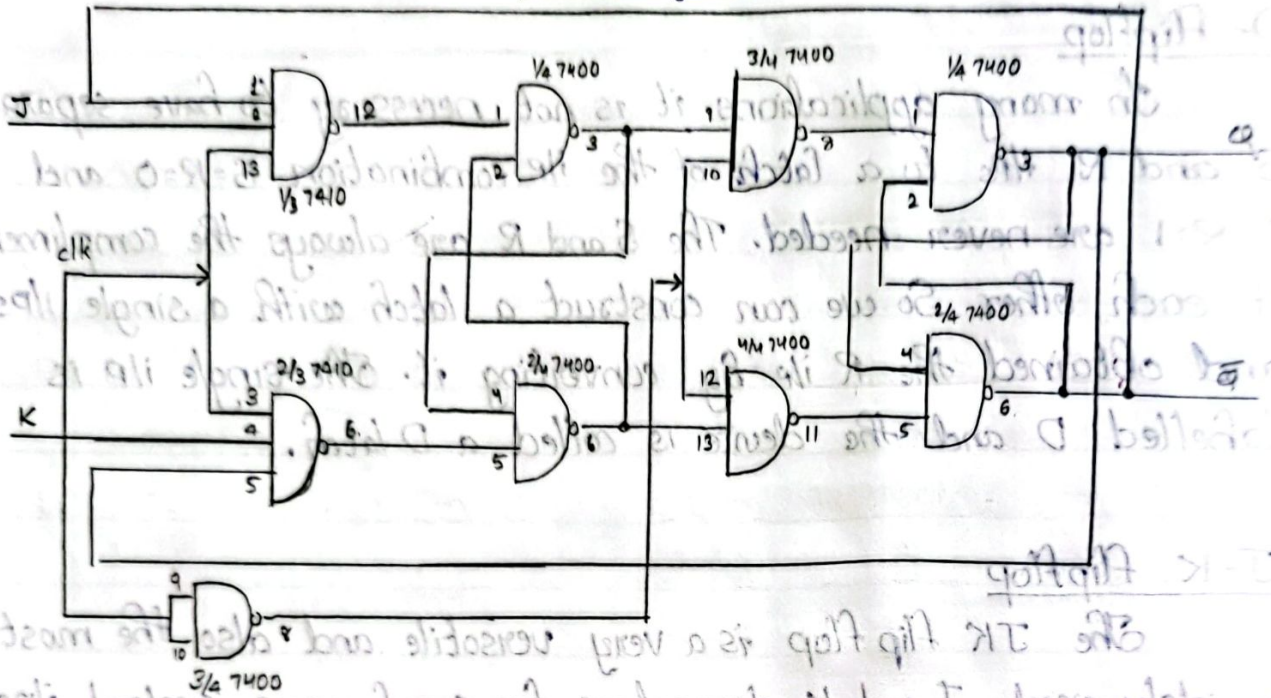
Result :

Realisation of flipflops are done and truth table verified.

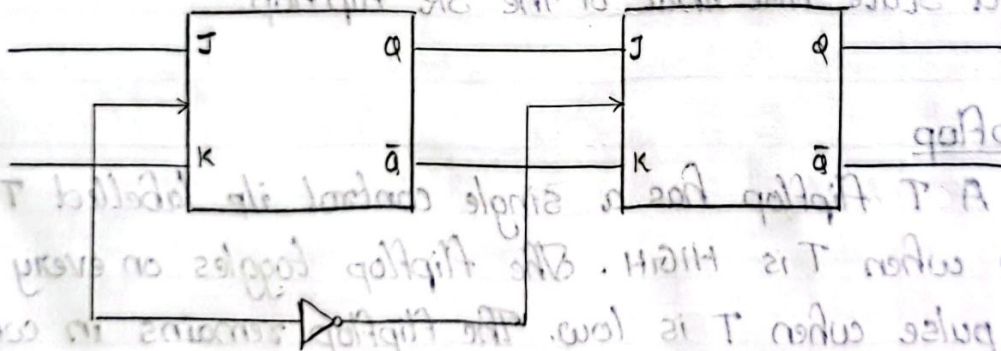
1	0	1	0	0
1	0	1	0	10
1	0	1	0	11
1	0	1	0	01

Master Slave JK Flip Flop

Circuit Diagram



Logic Symbol



J	K	Q _n	State
0	0	Q _n	No change
0	1	0	Reset
1	0	1	Set
1	1	Q _n	Toggle